The Derived Sign: Local Clausius Gives Einstein+Λ, and Einstein+Λ Gives the Cosmological Minus Sign

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Abstract

This is the derivation that exists. It is not Faulkner–Guica–Hartman– Myers–Van Raamsdonk on a net of de Sitter balls, and it is not a Nobel theorem.

Locally: the Unruh temperature is positive, $\delta Q = T\,dS$ on a local Rindler horizon with $dS = \eta\,\delta A$ implies Einstein's equation with an undetermined Λ [**?**]. That algebra is **[P]** (Paper 18). The spinless vacuum with $\Lambda > 0$ is de Sitter (Paper 19). Torsion is not in the problem.

Globally, on the cosmological horizon of Schwarzschild–de Sitter, the same field equation implies

$$T\,\mathrm{d}S + \mathrm{d}M = 0.$$

At $M = 0$, $\mathrm{d}r_c/\mathrm{d}M = -1$ and $T\,\mathrm{d}S = -\mathrm{d}M$. The minus sign is *derived* from Einstein+Λ, not copied from AdS. **[P]** (implicit differentiation, audited; finite-M residual $< 10^{-6}$).

Copying the Casini–Huerta–Myers plus sign onto the Gibbons–Hawking horizon remains false (Paper 17). Using the minus sign on that horizon is using Einstein+Λ. The two signs live on two surfaces and are consistent.

What is not **[P]**: a holographic pair, a value of Λ, FGHMV equivalence for arbitrary balls in de Sitter.

1 Local implication (cited, already isolated)

Unruh: $T = \hbar\kappa/2\pi > 0$ for a boost. Clausius on a local Rindler horizon with $\theta = \sigma = 0$ at the bifurcation surface, together with $dS = \eta\,\delta A$ and $\nabla^a T_{ab} = 0$, implies [**?**, **?**]

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{2\pi}{\hbar\eta}\,T_{\mu\nu}, \qquad \Lambda \text{ undetermined.} \tag{1}$$

The Unruh sign is plus. It is a theorem of QFT, not an AdS import. The area law is an assumption. **[P]** as an implication under those hypotheses.

2 Vacuum

If $T_{\mu\nu} = 0$ and $\Lambda > 0$, (**??**) is de Sitter of radius $\ell = \sqrt{3/\Lambda}$ (Paper 19). In the flat FLRW chart, $a = e^{Ht}$ with $H^2 = \Lambda/3$ saturates both Friedmann equations. Torsion vanishes with the spin. **[P]**

3 The cosmological sign, derived

Schwarzschild–de Sitter ($G = c = \hbar = 1$)

$$f(r) = 1 - \frac{2M}{r} - \frac{r^2}{\ell^2}, \qquad \Lambda = \frac{3}{\ell^2}. \tag{2}$$

The cosmological root $r_c(M)$ of $f(r) = 0$ satisfies $r_c(0) = \ell$. Implicit differentiation,

$$\frac{\mathrm{d}r_c}{\mathrm{d}M} = -\frac{\partial_M f}{\partial_r f} = \frac{\ell^2 r}{M\ell^2 - r^3}, \qquad \left.\frac{\mathrm{d}r_c}{\mathrm{d}M}\right|_{M=0} = -1. \tag{3}$$

Surface gravity $\kappa = |f'(r_c)|/2$ equals $1/\ell$ at $M = 0$, so $T = \kappa/2\pi = 1/(2\pi\ell)$. Entropy $S = \pi r_c^2$ then gives

$$T\,\mathrm{d}S = \frac{1}{2\pi\ell} \cdot 2\pi\ell \cdot \mathrm{d}r_c = -\mathrm{d}M \qquad (M = 0). \tag{4}$$

The first law of the cosmological horizon is $T\,\mathrm{d}S + \mathrm{d}M = 0$. The auditor recovers (**??**)–(**??**) identically in sympy. At finite small M the same combination is zero to 10^{-6}. Without the Λ term there is no cosmological root. **[P]**

This is the Gibbons–Hawking minus sign [**?**, **?**], now as an identity of (**??**), not a citation standing alone.

The physical-process reading is the same identity: positive Killing energy crossing the past horizon focuses the generators (Raychaudhuri plus Einstein), so the area at the bifurcation surface falls [**?**].

4 Two surfaces, two signs, no contradiction

Surface	Sign	Source
Local Rindler / CHM ball	$+\delta\langle T_{00}\rangle \Rightarrow +\delta S$	Unruh + Clausius
Cosmological horizon	$+\mathrm{d}M \Rightarrow -\mathrm{d}S$	Einstein+Λ on SdS, (**??**)

Putting the first sign on the second surface is false (Paper 17). Putting the second sign on the second surface is Einstein+Λ. The derivation does not copy AdS onto de Sitter. It uses Unruh locally and (**??**) globally.

5 What this derivation is

1. Local entanglement/Clausius with the Unruh plus sign $\Rightarrow$ Einstein+Λ (hypotheses of [**?**]).
2. Vacuum of that equation with $\Lambda > 0 \Rightarrow$ de Sitter.
3. The same equation on SdS $\Rightarrow$ cosmological $T\,\mathrm{d}S + \mathrm{d}M = 0$.

That is emergent bookkeeping yielding de Sitter *as the vacuum of the metric sector it produces*, with the global sign derived. It is not an FGHMV theorem about a cosmological CFT, and it does not fix Λ.

Admission. *I derived the cosmological minus sign from Schwarzschild–de Sitter and recorded the local implication that was already isolated. I did not produce a holographic pair, a net of CHM balls in de Sitter, or a reason to award a prize. The vacuum is de Sitter. The first law that selected Einstein+Λ is still local Clausius plus an area law, not FGHMV in cosmology.*

References

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